TEACHING PORTFOLIO | SAMPLE SYLLABUS

Conservation Biology & Sustainable Development

Field, laboratory, quantitative, and discussion-based approaches to biodiversity in human landscapes.

Natalia Borrego | Portfolio adaptation of ENV/BIO 3009/4900, taught at Baruch College, CUNY (2018).

Portfolio note. This version preserves the course's original interdisciplinary and field/laboratory design while updating learning outcomes, organization, and assessment language. Institution-specific dates, policies, and logistics are omitted.

COURSE DESCRIPTION

Conservation biology asks how biological diversity can be measured, why it changes, and what kinds of evidence should guide decisions about species, populations, ecosystems, and landscapes. Because conservation and resource management occur in human landscapes, the course integrates ecological, social, legal, and cultural dimensions rather than treating biological problems as isolated from people.

Students work across scales - from genes and populations to ecosystems and regions - using field observations, quantitative exercises, primary literature, case studies, and practitioner perspectives.

LEARNING OUTCOMES

1. Quantify and compare biological diversity using appropriate measures, and explain what each measure captures or obscures.
2. Analyze how ecological and human processes contribute to biodiversity loss across biological scales.
3. Use ecological data and models to evaluate conservation or management alternatives.
4. Interpret primary literature and distinguish evidence, assumptions, uncertainty, and inference.
5. Integrate ecological evidence with social, legal, and cultural context when evaluating decisions.
6. Communicate an evidence-based recommendation in written, oral, and visual forms.

COURSE ARCHITECTURE

1 | Foundations

What is biodiversity? How do we measure it? Why does scale matter?

2 | Change & threats

Habitat change, exploitation, invasive species, climate change, genetic consequences.

3 | Tools & scales

Population viability, genetics, spatial thinking, community and ecosystem approaches.

4 | Decisions

Protected areas, restoration, reintroduction, sustainable use, evidence-based tradeoffs.

HOW STUDENTS WORK

Field observation
Local biodiversity and conservation issues become data and cases rather than abstract examples.

Quantitative ecology
R, spreadsheet exercises, population models, and spatial tools connect concepts to inference.

Primary literature
Students read papers for question, method, evidence, assumptions, and conclusion.

Case discussion
Students evaluate competing management options and articulate what evidence would change their judgment.

Practitioner perspectives
Guest experts and site-based activities connect academic concepts to professional decisions.

Student research
Teams collect or analyze data and communicate the question, analysis, result, and limits of inference.

ASSESSMENT DESIGN

Assessment emphasizes application, analysis, and communication.

The original course combined exams, homework/classwork, writing, and an optional research project for advanced enrollment. In this portfolio adaptation, those components are organized around the scientific work students are expected to perform.

Component	Purpose	Examples
Preparation & in-class analysis	Frequent practice before higher-stakes synthesis.	Reading responses, short data exercises, case discussions, brief presentations.
Quantitative / field exercises	Connect ecological concepts to measurement and inference.	Biodiversity metrics, R or spreadsheet analysis, GIS/spatial exercises, population viability analysis.
Synthesis assessments	Evaluate whether students can connect mechanisms across scales.	Short-answer and data-interpretation questions; cumulative conceptual synthesis.
Writing assignment	Develop evidence-based scientific argument and communication.	Analysis of a conservation problem using literature and explicit treatment of uncertainty.
Collaborative research project	Move from following procedures to making research decisions.	Question, data collection or analysis, interpretation, presentation, and group paper.

INCLUSIVE COURSE DESIGN

Lower the cost of access

The original course was a zero-textbook course using freely available resources. A current implementation would retain open or library-accessible readings wherever possible.

Multiple routes into participation

Discussion, written preparation, data analysis, field observation, small-group work, and visual communication allow students to demonstrate reasoning in more than one format.

Transparent scientific tasks

Assignments specify the question students are answering, the evidence they should use, and the inference they are being asked to make.

Local context, transferable concepts

Field and practitioner activities are adapted to the institution's region so students learn general ecological principles through systems they can directly observe.

Teaching intent. Students should leave the course able to do more than name threats or management tools. They should be able to ask what evidence a decision requires, determine whether available data support the proposed inference, and explain how ecological and human context alter the answer.

REPRESENTATIVE COURSE SEQUENCE

Module	Core question	Representative activities and tools
Foundations	**What is biodiversity, and what does it mean to measure it?**	Biodiversity metrics; reading scientific papers; introductory R/data work.
Human dimensions	**Why are conservation problems also social problems?**	Environmental justice; ecological economics; practitioner discussion.
Habitat & landscape change	**How does spatial structure alter ecological processes?**	GIS / Google Earth; fragmentation exercises; spatial interpretation.
Exploitation & invasions	**How do population processes shape sustainable use and biological invasions?**	Harvest models; applied demography; case analysis.
Climate & genetics	**How does environmental change interact with genetic variation?**	Conservation genetics; student presentations on emerging threats.
Population & species management	**What evidence is needed to decide whether a population is viable?**	Population viability analysis; matrix models; management scenarios.
Protected areas & restoration	**When do protection, reintroduction, or restoration strategies work?**	Case studies; reintroduction evidence; protected-area design.
Sustainable use & synthesis	**How should ecological evidence inform decisions under competing goals?**	Marine spatial planning / resource allocation; group research presentations.

SAMPLE RESEARCH PROJECT SCAFFOLD

From a conservation problem to a defensible inference.

Question
Define a tractable ecological or management problem and identify what must be measured.

Evidence
Collect or assemble data and document assumptions behind the measurements.

Analysis
Choose an analysis that matches the question and evaluate alternative interpretations.

Communication
Present the result, uncertainty, limits, and implications in a short talk and written product.

MATERIALS & TEACHING MODES

The historical course used a freely available conservation biology text, open spreadsheet exercises, primary literature, podcasts, and web-based instructional modules. A current version would continue to prioritize open or library-accessible materials and update readings and tools to match the local field context and current literature.

Short lecture
Conceptual foundations needed for active work.

Discussion
Competing explanations, assumptions, management tradeoffs.

Laboratory analysis
Data, models, and quantitative tools.

Field learning
Measurement connected to organisms and landscapes.

Collaborative research
Shared responsibility for data, interpretation, communication.

Independent writing
Construct and defend an evidence-based argument.